

Day 19: Advanced & Specialized Visualizations

📌 Definition to Remember

Advanced visualizations unlock deeper exploratory analysis through multi-axis correlation (**Combo Charts**), variance breakdown (**Waterfall**), AI-powered root-cause discovery (**Decomposition Tree**), and pipeline conversion (**Funnel Charts**).

1. Specialized Visual Types

1. Line and Clustered Column (Combo Chart):

- Displays volume bars (Sales Revenue) on Primary Y-Axis and margin % line on Secondary Y-Axis.

2. Waterfall Chart (Variance Walk):

- Breaks down financial variances from Budget to Actuals, or Prior Year to Current Year showing positive/negative bridge steps.

3. Decomposition Tree (AI Visual):

- Allows users to dynamically drill down into root-cause dimensions (e.g., Drill into Revenue by Region → Category → Salesperson).

4. Scatter / Bubble Plot:

- Visualizes correlations across 3 dimensions (**X**-Axis, **Y**-Axis, and Bubble Size) with anomaly clustering.

5. Funnel Chart:

- Tracks stage-by-stage drop-off in conversion pipelines (e.g., Lead → Qualified → Demo → Closed Won).

6. Geographic Maps:

- Filled Maps (Choropleth), Bubble Maps, and Azure Map layers with geographic coordinates.

★ Must-Master Skills & Interview Points

1. How to configure Secondary Y-Axis scales on combo charts without misleading readers.
2. Use cases for Waterfall charts in financial quarterly earnings reports.
3. Leveraging the Decomposition Tree's "High Value / Low Value" AI explainers.

⚡ Quick Recall

Combo (Dual Axis) + Waterfall (Variance Walk) + Decomposition Tree (Root Cause AI) + Funnel (Conversion)